

Seminar Logic 13

Multimea numerelor naturale

Axiomele lui Peano:

1) $\exists 0$

2) dacă $\exists m$ at $\exists s(m)$ ^{↑ succesorul lui m}
 $\left(\begin{smallmatrix} 1 \\ m+1 \end{smallmatrix} \right)$

$$\mathbb{N} = \left\{ \begin{array}{l} \emptyset = 0, \\ \{\emptyset\} = 1, \\ \{\emptyset, \{\emptyset\}\} = 2, \\ \{\emptyset, \{\emptyset, \{\emptyset\}\}, \{\emptyset, \{\emptyset, \{\emptyset, \{\emptyset\}\}\}\} = 3, \dots \end{array} \right.$$

Operații cu nr nat:

Adunarea nr nat. Def:

1) $m + 0 \stackrel{\text{def}}{=} m$

2) $m + s(n) \stackrel{\text{def}}{=} s(m+n)$

$0 + 0 \stackrel{d_1}{=} 0$

$1 + 0 \stackrel{d_1}{=} 1$

$2 + 0 \stackrel{d_1}{=} 2$

$0 + 1 \stackrel{d_2}{=} 0 + s(0) \stackrel{d_2}{=} s(0+0) = s(0) = 1$

$1 + 1 = 1 + s(0) = s(1+0) = s(1) = 2$

$\vdots$
 $(m+1) \stackrel{d_4}{=} m + s(0) = s(m+0) \stackrel{d_4}{=} s(m)$

$0 + 2 = 0 + s(1) = s(0+1) = s(1) = 2$

$\vdots$

Examen Def: ad nr nat, nr int, dem orice, inmult, nr nat

Proprietăți ad nr nat. Fie $m, n, p \in \mathbb{N}$

① Asociativitate: $(m+n)+p = m+(n+p)$

Principiul inducției nat.

I $P(0)$ ade

II $P(k) \rightarrow P(k+1) \Leftrightarrow P(k) \rightarrow P(s(k))$

Dem prim ind. mat. $\text{dup} p = m \cdot \text{mat}(\text{degem cel mai din stnga})$

$$P(p) : (m+n) + p = m + (n+p) \quad \forall m, n \in \mathbb{N}$$

$$I \quad P(0) : (m+n) + 0 \stackrel{?}{=} m + (n+0)$$

$$\Rightarrow (m+n) + 0 \stackrel{d1}{=} m+n$$

$$\Rightarrow m + (n+0) \stackrel{d1}{=} m+n \quad \left. \vphantom{\begin{matrix} \Rightarrow (m+n) + 0 \stackrel{d1}{=} m+n \\ \Rightarrow m + (n+0) \stackrel{d1}{=} m+n \end{matrix}} \right\} \Rightarrow P(0) \text{ ade}$$

$$II \quad P(k) \rightarrow P(k+1) \Leftrightarrow P(k) \rightarrow P(\sigma(k))$$

$$P(k) : (m+n) + k = m + (n+k) \text{ ade}$$

$$\text{Dem} = P(\sigma(k))$$

$$(m+n) + \sigma(k) = m + (n + \sigma(k))$$

$$\Rightarrow (m+n) + \sigma(k) \stackrel{d2}{=} \sigma((m+n) + k)$$

$$\Rightarrow m + (n + \sigma(k)) = m + \sigma(m + (n+k)) = \sigma(m + (n+k))$$

$$\bullet (m+n) + \sigma(k) = \sigma((m+n) + k)$$

$$\bullet m + (n + \sigma(k)) = m + \sigma(m + (n+k)) = \sigma(m + (n+k)) \Rightarrow$$

$$P(k) \text{ ade}$$

$$\Rightarrow \sigma((m+n) + k) = \sigma(m + (n+k)) \text{ ade}$$

$$\Rightarrow P(p) \text{ ade } \forall p \in \mathbb{N}$$

$$(2) \text{ „0 este elem neutru”} : m+0 = \underline{0+m} = m$$

$$P(m) : 0+m = m \quad \forall m \in \mathbb{N}$$

Conform def 1) $\Rightarrow m+0 = m$, dem (c) $0+m = m$

$$\text{Dem } P(m) : 0+m = m \quad \forall m \in \mathbb{N}$$

$$I \quad P(0) : 0+0 = 0 \stackrel{d1}{=} 0 = 0 \text{ ade}$$

$$II \quad P(k) \rightarrow P(\sigma(k))$$

$$P(k) : 0+k = k \text{ ade}$$

$$\text{Dem } P(\sigma(k)) \Rightarrow$$

$$\Rightarrow 0 + \sigma(k) \stackrel{?}{=} 0$$

$$\stackrel{d2}{=} \sigma(0+k) = 0$$

$$\stackrel{d1}{=} \sigma(k) = 0 \text{ ade}$$

③ Comutativitatea $m+n = n+m$

Dem prin inducție după $m \in \mathbb{N} : P(m) : m+n = n+m$

I $P(0) : 0+n = n+0$ adică pt că 0 elem neutru

II $P(k) \rightarrow P(s(k))$

$P(k) : 0+k = k+0$ adică

$P(s(k)) : s(k)+n = n+s(k)$

$$n+s(k) \stackrel{d_2}{=} s(n+k) = (k+1)+n = k+(1+n)$$

$$s(k)+n \stackrel{\text{vezi succesorul toate adunăturilor}}{=} (n+k)+1 \stackrel{ip}{=} (k+n)+1 = k+(n+1) \stackrel{d_1}{=} k+(1+n)$$

$$k+(1+n) = n+s(k)$$

$\Rightarrow P(s(k))$ adică

$\rightarrow P(m)$ adică

5) Dacă $m+p = n+p$ atunci $m=n$

Dem prin inducție după $p \in \mathbb{N}$ (Term)

6) Dacă $m+n = 0$ atunci $\underset{\in \mathbb{N}}{m} = \underset{\in \mathbb{N}}{n} = 0$

(MRA) $\rightarrow$ Euclid pt nr prime
prime dem dem matematică

(MRA) $\forall p \in \mathbb{N} : m \neq 0 \Rightarrow \exists p \in \mathbb{N} : s(p) = m$

$$m+n=0 \Leftrightarrow s(p)+n=0 \stackrel{Com}{\Leftrightarrow} m+s(p)=0 \stackrel{d_2}{\Leftrightarrow}$$

$$\stackrel{d_3}{\Leftrightarrow} s(\underbrace{m+p}_{\in \mathbb{N}}) = 0 \quad \swarrow \Rightarrow \boxed{m=0} \quad \downarrow \Rightarrow$$

$$m+n=0$$

$$\Rightarrow 0+n=0 \stackrel{Com}{\Leftrightarrow} n+0=0 \stackrel{d_1}{\Leftrightarrow} \boxed{n=0}$$

7) Trichotomie: Dem wenn 3 Alternativen die bei exactly one (i) (ii) (iii)

(i) $m = n$

(ii) $\exists p \in \mathbb{N}^+ \text{ or } m + p = n \rightarrow \text{rel} >$

(iii) $\exists p \in \mathbb{N}^+ \text{ or } m + p = n \rightarrow \text{rel} <$

(Lemma)

Induktions zur nat. Def:

1) $m \cdot 0 \stackrel{\text{def}}{=} 0$

2) $m \cdot \alpha(m) \stackrel{\text{def}}{=} m \cdot m + m$

Commutativität: $m \cdot n = n \cdot m$

! denn diste $m \cdot k + m = k \cdot m + m$